Explicit High-Level Semantic Network for Domain Generalization in Hyperspectral Image Classification

Xusheng Wang, Shoubin Dong, *Member, IEEE*, Xiaorou Zheng, *Member, IEEE*, Runuo Lu, and Jianxin Jia

***Abstract*— When applied across different scenes, hyperspectral image (HSI) classification models often struggle to generalize due to the data distribution disparities and labels' scarcity, leading to domain shift (DS) problems. Recently, the high-level semantics from text has demonstrated the potential to address the DS problem, by improving the generalization capability of image encoders through aligning image–text pairs. However, the main challenge still lies in crafting appropriate texts that accurately represent the intricate interrelationships and the fragmented nature of land cover in HSIs and effectively extracting spectral–spatial features from HSI data. This article proposes a domain generalization (DG) method, EHSnet, to address these issues by leveraging multilayered explicit high-level semantic (EHS) information from different types of texts to provide precisely relevant semantic information for the image encoder. A multilayered EHS information paradigm is well-defined, aiming to extract the HSI's intricate interrelationships and the fragmented land-cover features, and a dual-residual encoder connected by a 2-D convolution is designed, which combines CNNs with residual structure and Vision Transformers (ViTs) with short-range cross-layer connections to explore the spectral–spatial features of HSIs. By aligning text features with image features in the semantic space, EHSnet improves the representation capability of the image encoder and is endowed with zero-shot generalization ability for cross-scene tasks. Extensive experiments conducted on three hyperspectral datasets, including Houston, Pavia, and XS datasets, validate the effectiveness and superiority of EHSnet, with the Kappa coefficient improved by 8.17%, 3.22%, and 3.62% across three datasets compared to the state-of-the-art (SOTA) methods. The code is available at https://github.com/SCUT-CCNL/EHSnet.**

***Index Terms*— Domain generalization (DG), hyperspectral image (HSI) classification, multiple modality, visual–language model.**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of continuous and finely divided spectral bands, providing rich spatial and spectral information. They are widely applied in fine-grained land-cover classification [1], precision agriculture [2], [3], and environmental monitoring [4], [5]. Over the past decade, deep learning methods have made notable achievements in HSI classification [6], [7], [8], [9]. However, while most of them excel in the source domain, they tend to experience a significant drop in performance when transferred to an unknown target domain, leading to domain shift (DS) problems [10]. This is primarily attributed to the variations in spectral curve distributions within the same HSI category (Fig. 1) due to sensor discrepancies, weather conditions, geographic locations, and other factors. As a result, these interdomain disparities hinder many HSI classification methods from effectively generalizing in the target domain, severely limiting the progress of HSI classification models for large-scale applications.

Due to the high cost of obtaining data [1] and the challenges involved in collecting labels [11], HSIs and corresponding labels are relatively scarce. This scarcity poses difficulties in addressing the DS problem of HSIs. In recent years, many studies have addressed the DS problem through transfer learning [12], [13], [14], [15]. Among them, domain generalization (DG) has emerged as a promising solution, which only accesses data and labels from the source domain during training and directly transfers to the target domain [16]. DG can be categorized into single-source DG (SSDG) and multi-source DG (MSDG) based on the number of source domains. For instance, Dong et al. [17] proposed a spatial–spectral enhancement method with causal constraint for SSDG with the objective of accurate causal alignment and efficient data augmentation. Qi et al. [18] proposed a two-branch network for MSDG to explore the internal attributes of each source domain and generic features among different source domains. SSDG, which is the focus of this study, is more general and challenging than MSDG [19], as it involves limited information from only one source domain, increasing the difficulty of transferring to the target domain. When dealing with sparse data distribution in high-dimensional HSIs, DG methods typically focus on simulating the diversity of the target domain through data augmentation to support the

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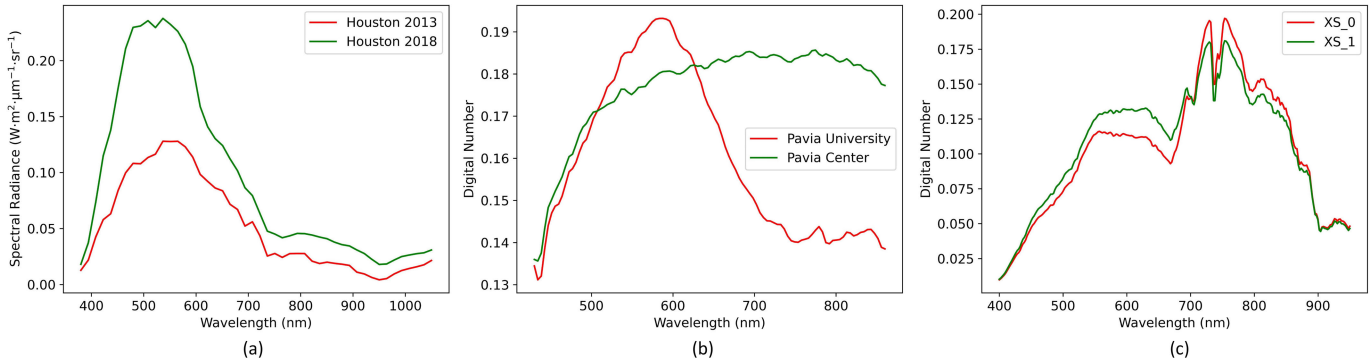


Fig. 1. Average digital number or spectral radiance curves of the source and target dataset across three datasets, with one class selected as an example for each. (a) Class water of the Houston dataset with Houston 2013 as the source domain and Houston 2018 as the target domain. (b) Class bitumen of the Pavia dataset with Pavia University as the source domain and Pavia center as the target domain. (c) Class farmland of the XS dataset with XS_0 as the source domain and XS_1 as the target domain. The digital numbers have been normalized to the same range.

model in extracting domain-invariant features. However, relying on the information derived from the transformations of the source data is limited, as it cannot introduce new features and fails to fundamentally resolve the problem of insufficient representation capability.

Despite some advancements made by DG or DS models, these models are often trained only on vision-modality annotations and tasks, lacking alignment with human cognition. This leads to poor performance when confronted with data distributions that differ from the training set. In contrast, text modalities are inherently rich in supervisory information. Language models, particularly large language models, can capture a significant amount of high-level cognition from text modalities that align with human understanding, referred to as explicit high-level semantics (EHS). Numerous studies have demonstrated the effectiveness of integrating EHS into visual tasks [20], [21], [22]. These methods, including large visual–language pretrained models, supplement EHS information from text paired with images, enabling the image encoder to learn more high-level feature representations, which plays an important role in downstream tasks [23]. Radford et al. [24] proposed CLIP, utilizing natural language supervision to achieve exceptional generalization by pretraining on a substantial dataset containing 400 million images with matched texts. Inspired by this, many visual–language methods have emerged for remote sensing (RS). For example, Li et al. [25] introduced a visual–language model for RS image classification on the basis of contrastive learning between vision and language. These studies mainly focus on natural images or multispectral images, with less emphasis on HSIs. Zhang et al. [10] proposed to use coarse-grained text generated by prompt template and fine-grained text with detailed descriptions of land cover for cross-scene HSI classification. However, such fine-grained text is difficult to obtain in most hyperspectral datasets.

However, these visual–language models with EHS information encounter many difficulties in processing HSI images. First, it is difficult to design concise and effective EHS texts that can accurately provide rich high-level semantic information to represent the intricate interrelationships and the fragmented nature of land cover in HSIs. Second, it is necessary to improve the existing EHS models' visual encoders to ensure they can effectively capture the unique spatial–spectral information of high-dimensional HSIs.

To address the above issues, this article proposes an EHS network (EHSnet) for cross-scene HSI classification. EHSnet aligns image features and language features in a unified semantic space to assist the model's visual representation learning through multilayered high-level semantic information. Specifically, EHSnet consists of the following components: a hybrid image encoder for HSIs, a text encoder based on CLIP [24], and visual–language alignment. EHSnet's image encoder extracts the rich spatial and spectral features of HSIs through the serial combination of residual-style CNN modules and ViT modules with cross-layer connections. In the language part, EHSnet designs a multilayered EHS information paradigm, including global and local semantics, by obtaining two types of text using prompt templates, which are then encoded and integrated. Finally, by unifying visual modal information and language modal information in the semantic space through supervised contrastive learning, EHSnet has obtained strong representation capability for HSIs, enabling deep exploration of domain-invariant features and demonstrating good generalization in the target domain. The main contributions of EHSnet are summarized as follows.

1) We propose a DG method, EHSnet, with zero-shot generalization ability for cross-scene HSI classification by leveraging multilayered EHS information from texts to offer precisely relevant semantic information to the image encoder.
2) We define a multilayered EHS paradigm, including global and local semantics, to extract the intricate interrelationships and fragmented features of land cover in HSIs. Global semantics explore the full-image descriptive text annotated at the image level, while local semantics mine the local descriptive text corresponding to each patch.
3) We propose a dual-residual hybrid image encoder that combines a residual-structured 3-D-CNN network with a 2-D-CNN to distinguish the spatial information within different spectral bands, followed by a

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residual-structured ViT with short-range cross-layer connections that skip only one layer to fully explore the rich spectral–spatial features of HSIs.

4) We demonstrate that EHSnet has achieved state-of-the-art (SOTA) performance in cross-scene HSI classification by conducting comparative experiments with the DG methods for natural images and the SOTA methods for cross-scene HSI classification.

The remaining sections of this article are structured as follows. Section II introduces the relevant work of DG methods and visual–language models. Section III provides a detailed explanation of the proposed EHSnet. Section IV displays the experimental results and discussion. Finally, Section V concludes with conclusions and expectations.

II. Related Work

A. Domain Generalization

The DG methods aim to address the challenges posed by the DS problem and the lack of annotated labels in the target domain [26]. Existing DG methods are primarily categorized into three groups: data processing, representation learning, and learning strategies.

Data processing involves training the model by enhancing or generating out-of-domain sample data related to the source domain. Li et al. [27] proposed a progressive domain expansion network (PDEN), which creates additional domains to simulate the changes in photometry and geometry across unknown domains. Cheng et al. [28] proposed adversarial Bayesian augmentation (ABA) to assist the model in generalizing to unseen domains through data augmentations.

Domain-adversarial training is commonly utilized in representation learning to acquire domain-invariant representations. Zhao et al. [29] proposed a locally linear unbiased randomization network (LLURnet) using style randomization to build a generative network, along with adversarial learning for gaining domain-invariant features. Zhang et al. [30] proposed a single-source domain expansion network (SDEnet) to extract the domain-invariant features through an adversarial contrastive method.

Learning strategies refer to utilizing mature learning models to improve model generalization capability. Nam et al. [31] proposed style-agnostic networks (SagNets) to reduce the inherent style bias of CNNs to narrow the gap between domains and focus more on the contents. Chen et al. [32] proposed a meta-causal learning method to infer the cause of DS between the source domain and the auxiliary domain, which is established to simulate the target domain. Zhang et al. [33] proposed a new federated DG method with generalization adjustment (FedDG-DA) to guarantee the generalization under DS.

However, when handling HSI data with a sparse distribution, most DG methods that depend on information from source data transformations or augmentations struggle to fundamentally enhance the model's deficient representation capacity.

B. Visual–Language Model

Compared to models trained solely under image supervision, visual–language models leverage language supervision with images, showing a strong capability for visual representation learning. The concept of projecting images and texts into a shared embedding space adopted by large visual–language pretraining models such as CLIP [24], ALIGN [34], and so on has been researched for almost a decade [35], [36], [37]. However, due to the contrastive learning methods, extensive training datasets, and text representation learning with transformers, large visual–language models have shown strong generalization capabilities in downstream tasks.

Although pretrained visual–language models have learned a wide range of feature representations, their adaptation to specialized domains like RS remains challenging. Therefore, many visual–language models adapted to RS have been developed [38], [39], [40], [41]. As for HSI, visual–language research is quite scarce. LDGnet [10] proposed to use coarse and fine-grained texts for each class, enhancing cross-scene HSI classification performance by image–text aligning. Fine-grained texts need to be designed by manually observing the land-cover conditions of each class and combining prior knowledge such as color, shape, distribution, and adjacency relationships [10]. However, such fine-grained texts are difficult to obtain in most hyperspectral datasets due to the intricate interrelationships and the fragmented nature of land cover in HSIs, especially in large hyperspectral datasets covering large areas. In addition, the land-cover conditions within the same class may also vary greatly because of the changes in time and space between domains, and the fine-grained texts may be inapplicable or even conflicting across domains.

Most of the current visual–language methods are designed to process natural images, lacking the capacity to efficiently extract the inherent spatial–spectral information of HSIs. At the same time, facing the intricate interrelationships and the fragmented nature of ground objects in HSIs, designing the textual modal information is also a challenging problem that needs to be solved.

III. Proposed Method

The flowchart of the proposed EHSnet can be seen in Fig. 2. A patch of size $13 \times 13 \times$ bands is sampled from the source domain and then fed into the image encoder to learn the unique features of HSI data and calculate the classification loss, denoted by Loss_{cls}. The corresponding text of the sampled patch enters the text encoder together with the global text corresponding to the source domain. After the two types of texts are encoded by the text encoder and fused by a parameter, they enter the semantic space with the image features for supervised contrastive learning to obtain the $\text{Loss}_{\text{clip}}$. In the testing stage, the prediction results for the target domain are given only through the CLS head of the image encoder. In this section, let $X = \{x_i\}_{i=1}^{N} \in \mathbb{R}^{\text{bands}}$ represent the data from the source domain, and the corresponding class labels are represented as $Y = \{y_i\}_{i=1}^{N}$, where bands and N represent the data dimensions and the number of samples from the source domain, individually.

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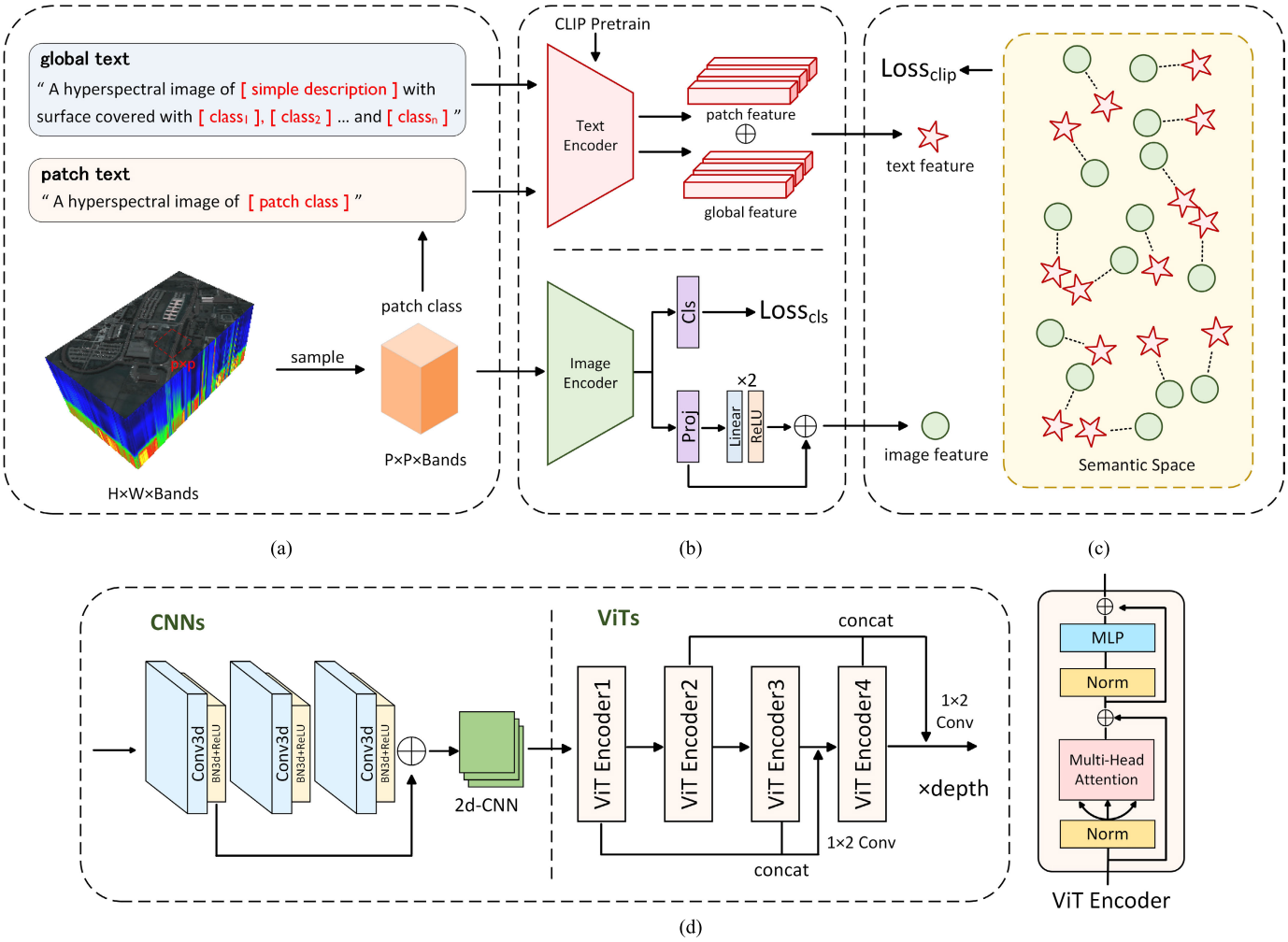


Fig. 2. Illustration of the proposed EHSnet. HSI patches and their respective patch texts and global text corresponding to the source domain are inputted into the model to extract visual and textual features. Two text features are fused and then aligned with the visual features in the semantic space. In the testing phase, predictions are generated from the CLS head of the visual encoder. (a) Image patch and text preparation. (b) Feature extraction. (c) Visual–linguistic alignment. (d) Image encoder.

A. Image Encoder

HSIs refer to 3-D data containing hundreds of spectral channels, rich in spatial and spectral information. Therefore, a residual 3-D-CNN architecture with three layers of depth is employed at the beginning of the image encoder. As shown in Fig. 2, the three layers of Conv3d are followed by BN3d-ReLU, respectively, and the output generated from the first layer is then further linked with the third layer in a residual manner. The residual 3-D-CNN architecture with three layers of depth can extract the spectral information of the hyperspectral data. After that, a 2-D-CNN layer is connected, which can strongly distinguish the spatial details within various spectral channels while preserving valuable spectral information. It is crucial for extracting features from HSI data [7].

The CNNs-based HSI classification method has greatly improved performance due to its excellent feature representation capability. However, CNNs struggle to establish dependencies between long sequence data and, therefore, have limited capabilities in processing HSI spectral sequence features and obtaining deep semantic features. To make up for this shortcoming, we introduce the Vision Transformer (ViT), leveraging the connection of the 2-D-CNN layer, to extract deeper semantic features of HSIs and adopt a short-range cross-layer connection design [42] that skips only one layer in the ViT modules to reduce the possibility of losing valuable information in the hierarchical propagation process. The combination of these two residual-style encoders provides powerful representation capabilities for the hybrid image encoder.

Subsequently, the visual features of HSI data are transmitted to both the CLS layer and the Proj layer. Loss_{cls} is calculated between the ground truth and the prediction given by the CLS head through the cross-entropy loss as

$$\text{Loss}_{\text{ce}}(p_i, y_i) = -\sum_{c} y_i^c \log\, p_i^c \tag{1}$$

y_i is the corresponding label class of x_i using one-hot encoding, c refers to the label index, and p_i represents the probability of the prediction generated through the CLS layer.

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Thus, Loss_{cls} is defined as

$$\text{Loss}_{\text{cls}} = \frac{1}{N}\sum_{i}^{N}\text{Loss}_{\text{ce}}(\text{Cls}(x_i), y_i). \tag{2}$$

N represents the number of training examples and the prediction of x_i given by the CLS head is denoted as $\text{Cls}(x_i)$.

After being encoded by the hybrid image encoder, image features will be projected through the Proj layer to maintain the same dimension as the text features and are then residual-connected with the outputs after two linear ReLUs to obtain more robust image features. Subsequently, the image features enter the semantic space for supervised contrastive learning with the text features.

B. Text Encoder

The text encoder in the proposed EHSnet consists of three layers and has eight attention heads with a parameter volume of about 33 M. It is built upon the Transformer architecture [43] and modified following the work of [44]. The Transformers in the text encoder utilize lower-cased byte pair encoding (BPE) with a vocabulary size of 49 152 [45] to encode the input text, which is also employed in CLIP. In addition, the maximum sequence length of the text encoder is limited to 77, as in CLIP. The encoded language features will be mapped to the semantic space to align with the visual features.

"An HSI of [patch class]" is designed as a prompt template in EHSnet to construct texts corresponding to each individual patch that represents local semantic information in a cloze form. However, the semantic information provided by the patch text generated from this prompt template is too limited to significantly enhance the model's representational capability. On the other hand, injecting a large amount of prior knowledge into text modal information, like LDGnet [10], to offer more semantic information requires a significant investment of manual effort. The intricate interrelationships and fragmented nature of land cover in HSIs make this even more difficult. Hence, a multilayered EHS paradigm is defined for HSIs. Specifically, a new prompt template of this paradigm is designed as follows: "An HSI of [simple description] with surface covered with $[\text{class}_1]$, $[\text{class}_2]$, and $[\text{class}_C]$" to construct the global text that represents global semantic information in a cloze form, where C represents the number of classes. This global text provides a concise overview of the relevant information and land-cover classes of the entire HSI. Since the same land-cover classes are shared between the source and target domains, this global semantic information applies to both. The two levels of text information complement each other, injecting more high-level semantic information into the model for extracting domain-invariant features. It is worth noting that we use prompt templates to generate two types of texts providing different levels of semantic information. This strategy conquers the challenges posed by the intricate interrelations and the fragmented nature of land cover in HSIs, facilitating scalability to diverse datasets. The global texts corresponding to the three datasets are displayed in Table I.

TABLE I
GLOBAL TEXT FOR THE THREE DATASETS

dataset	Global text
Houston	A hyperspectral image of the University of Houston with the surface covered with grass healthy, grass-stressed, trees, water, residential buildings, non-residential buildings and road
Pavia	A hyperspectral image of Pavia University with the surface covered with trees, asphalt, brick, bitumen, shadow, meadow, and bare soil
XS	A hyperspectral image of the suburb of Xiongan city with the surface covered with roads, buildings,trees, farmland, bare land, orchard and water

To align the multilayered EHS semantic information after obtaining the patch text features and global text features, a coefficient α is introduced to fuse the two text features in the form of (3) to generate the integrated text features f_{text}. These integrated features will be sent into the semantic space for supervised contrastive learning with the image features. The fused text feature contains multilayered semantic information and only needs to be aligned with the image features once, which saves the model's learning cost

$$f_{\text{text}} = (1-\alpha)\cdot f_{\text{patch}} + \alpha \cdot f_{\text{global}} \tag{3}$$

f_{patch} represents the patch text feature, while the global text feature is denoted as f_{global}.

C. Visual–Linguistic Alignment

After the visual feature and linguistic feature enter the semantic space, visual–language contrastive learning [46] is implemented to minimize the distance between the image feature and text feature belonging to the same class. The loss function of the supervised contrastive learning can be described in the following formula:

$$\text{Loss}_{\text{sup}} = -\sum_{i=0}^{N}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp\left(x_i^T x_p^+/\tau\right)}{\sum_{n\in N(i)}\exp\left(x_i^T x_n^-/\tau\right)}. \tag{4}$$

$P(i)$ and $N(i)$ denote the collections of the positive and negative instances, respectively, while x_p^+ and x_n^- represent an element from $P(i)$ and $N(i)$, respectively. The temperature parameter τ is used to govern the range of the logits in the softmax.

In the semantic space, the supervised contrastive loss of visual modality and linguistic modality, denoted as $\text{Loss}_{\text{clip}}$, is computed between the image–text pairs of the same class in a minibatch

$$\text{Loss}_{\text{clip}} = -\sum_{i=0}^{N}\frac{1}{|P(i)|}\left(\sum_{p\in P_v(i)}\log\frac{\exp\left(l_i^T v_p^+/\tau\right)}{\sum_{n\in N_v(i)}\exp\left(l_i^T v_n^-/\tau\right)} + \sum_{p\in P_l(i)}\log\frac{\exp\left(v_i^T l_p^+/\tau\right)}{\sum_{n\in N_l(i)}\exp\left(v_i^T l_n^-/\tau\right)}\right). \tag{5}$$

The collections of the positive and negative instances of the image features are denoted as $P_v(i)$ and $N_v(i)$, respectively, while $P_l(i)$ and $N_l(i)$ represent the collections of the positive

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and negative instances of the text features, respectively. And $|P(i)| = |P_v(i)| = |P_l(i)|$.

Due to the fused multilayered high-level semantic information, $\text{Loss}_{\text{clip}}$ can express not only the feature distance between the two modalities of the same class but also the distance between different classes. By optimizing $\text{Loss}_{\text{clip}}$, it can effectively reduce the feature distance of the two modalities of the same class while effectively expanding the distance of different classes. This integration of two levels of semantic information within the process of extracting spatial–spectral representation of HSI data aids the model in obtaining representations that are invariant across domains and specific to each class.

EHSnet uses both the text encoder and the image encoder for joint training in the source domain. In the inference stage, only the image data from the target domain is used, and the classification results are obtained through the CLS head. In EHSnet, the parameter β is introduced to balance the contribution of Loss_{cls} and $\text{Loss}_{\text{clip}}$, and, therefore, the total loss of EHSnet is represented as follows:

$$\text{Loss} = \text{Loss}_{\text{cls}} + \beta \cdot \text{Loss}_{\text{clip}} \tag{6}$$

IV. Experimental Results

In this section, three publicly available HSI datasets are first described. Then, the implementation details and SOTA methods for comparison are introduced. Next, extensive experiments and ablation analyses are conducted to assess the performance of the proposed EHSnet in HSI classification both quantitatively and qualitatively. Finally, we further discuss the performance of multilayer EHS in various aspects through a large number of comparative experiments and ablation studies.

A. Data Description

1) Houston Dataset: The dataset includes Houston 2013 [47] and Houston 2018 [48] scenes. The Houston 2013 dataset is 349 × 1905 in size and consists of 144 bands ranging from 364 to 1046 nm. The annotations for ground cover are divided into 15 classes. The Houston 2018 dataset is 2384 × 601 in size with 20 categories, containing 48 spectral bands. Experiments performed in this study use the shared 48 bands and seven classes with an overlapping area of 209 × 955 in size. The Houston 2013 dataset is chosen as the source domain, and the Houston 2018 dataset is chosen as the target domain. The sample number of the shared classes is displayed in Table II. Fig. 3 shows the pseudo-color and the ground-truth maps of the Houston dataset.

2) Pavia Dataset: The Pavia dataset comprises the Pavia University dataset and the Pavia Center dataset, both of which have a spatial resolution of 1.3 m and are divided into nine categories. The Pavia Center dataset is 1096 × 715 in size, containing 102 bands, while the Pavia University dataset is 610 × 340 in size, containing 103 spectral bands. The 102 shared bands and the seven shared classes between the scenes are used in the experiments. The Pavia University is chosen as the source domain, and the Pavia Center is chosen as the target domain. The sample number of the shared classes is displayed in Table III, and Fig. 4 shows the pseudo-color and the ground-truth maps of the Pavia dataset.

TABLE II
Number of Samples for the Houston Dataset

Class		The Number of Samples	
No.	Name	Houston 2013	Houston 2018
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-Residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

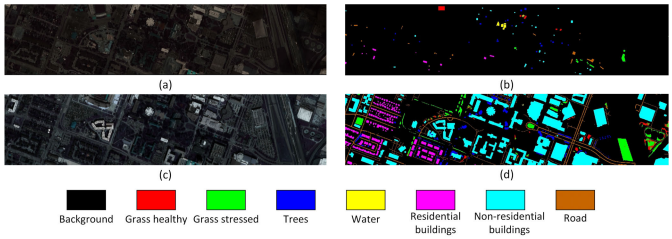


Fig. 3. (a) Visualization of Houston 2013 in pseudo-color. (b) Ground-truth visualization of Houston 2013. (c) Visualization of Houston 2018 in pseudo–color. (d) Ground-truth visualization of Houston 2018.

TABLE III
Number of Samples for the Pavia Dataset

Class		The Number of Samples	
No.	Name	Pavia University	Pavia Center
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

3) XS Dataset: The Xiongan Suburb (XS) dataset was collected in 2018 from the suburb of Xiongan City, Hebei Province, China [49], [50]. This dataset is 2821 × 2840 in size with a spatial resolution of 0.5 m and a spectral resolution of 2.4 nm, containing eight categories. Compared to the Pavia and Houston datasets, the XS dataset has a more fragmented category distribution, and the coarse label division leads to high heterogeneity within the class, which greatly increases the difficulty of classification. In the experiment, two nonadjacent subregions of the original image, each 1/16 the size of the original image, are used. The source domain is referred to as XS_0, and the target domain is referred to as XS_1, both of which are 706 × 710 in size and share the same seven classes and 253 bands. The sample number of the shared classes is displayed in Table IV, and Fig. 5 shows the pseudo-color and the ground truth maps.

B. Experimental Setup

1) Comparison With SOTA DG Methods: In this study, 8 DG methods are selected as comparison methods to validate the effectiveness of the proposed EHSnet. These consist of five SOTA DG methods applicable to natural images: PDEN [27], SagNet [31], Meta-casual [32],

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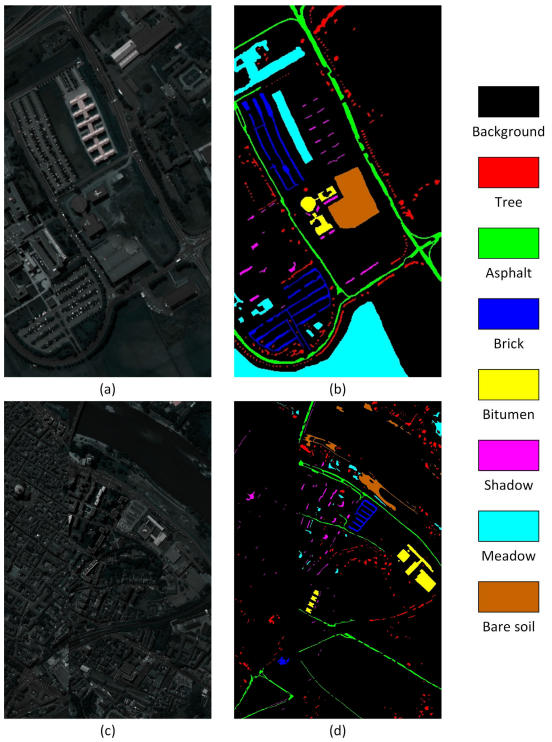


Fig. 4. (a) Visualization of Pavia University in pseudo-color. (b) Ground-truth visualization of Pavia University. (c) Visualization of Pavia Center in pseudo-color. (d) Ground-truth visualization of Pavia Center.

TABLE IV
NUMBER OF SAMPLES FOR THE XS DATASET

Class		The Number of Samples	
No.	Name	XS_0	XS_1
1	Road	31818	15744
2	Building	197170	155816
3	Tree	28376	59694
4	Farmland	85131	38547
5	Bare Land	115488	146609
6	Orchard	34658	57252
7	Water	2719	8454
Total		495360	482116

FedDG-DA [33], and ABA [28], and three SOTA DG methods for cross-scene HSI classification: SDEnet [30], LLURnet [29], and LDGnet [10].

1) PDEN. It is a progressive domain expansion network through generates supplementary domains to simulate the variations in photometry and geometry across unseen domains.
2) SagNet. It aims to reduce the inherent style bias of CNNs, thereby narrowing the gap between domains and shifting the focus more toward the content.
3) Meta-causal. This represents a learning method aimed at inferring the cause of DS between the source domain and the auxiliary domain, which is designed to simulate the target domain.
4) FedDG-DA. This is a novel federated DG method that incorporates generalization adjustment to ensure generalization in the presence of DS.

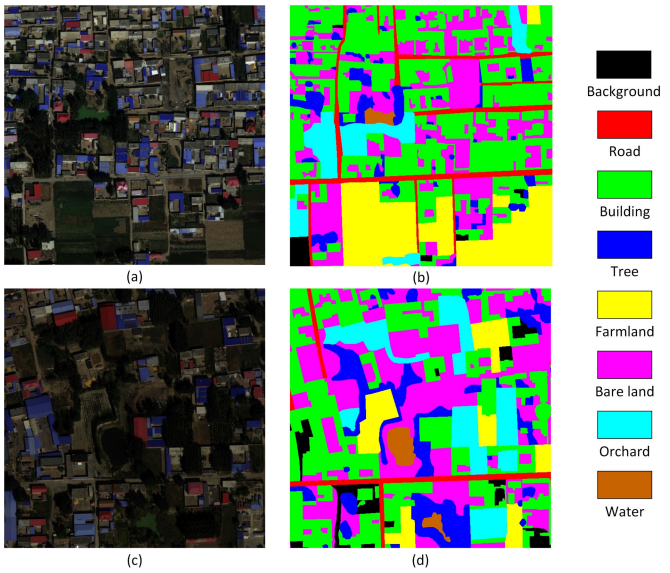


Fig. 5. (a) Visualization of XS_0 in pseudo-color. (b) Ground-truth visualization of XS_0. (c) Visualization of XS_1 in pseudo-color. (d) Ground-truth visualization of XS_1.

5) ABA. This method facilitates the model's generalization to unseen domains by utilizing adversarial Bayesian augmentation.
6) SDEnet. It constitutes a single-source domain expansion network designed to extract domain-invariant features using an adversarial contrastive method.
7) LLURNet. It utilizes style randomization to construct a generative network while also employing adversarial learning to acquire domain-invariant features.
8) LDGnet. It proposes the use of both coarse and fine-grained texts for each class, aiming to enhance the performance of cross-scene HSI classification through image–text alignment.

2) Evaluation Metrics: The classification performance of each model is assessed using three widely recognized indices: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). Furthermore, classification maps generated by different models are visualized to enable qualitative comparisons.

3) Implementation Details: In this study, the EHSnet is implemented on the PyTorch platform, utilizing the Adam optimizer, with a learning rate set to $1e-2$. Referring to previous work [10], [17], [29], [30], we set the patch size in the experiments to 13×13, which is also the optimal empirical parameter for patch size in previous studies. The batch size for all three datasets is set to 256. For the Houston and Pavia datasets, 80% of the data is used for training with 200 epochs. In addition, the Houston 2013 dataset is augmented by a factor of 5. While for the XS dataset, 10% of the data is used for training, and the model is trained for 100 epochs. EHSnet uses CLIP-ViT-B/32 as the initialization of the text encoder, and all experiments are completed with a single 24GB NVIDIA GeForce RTX 3090 Ti and an Intel[1] Xeon[1] Silver 4310 CPU.

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To ensure fairness, all methods are executed on the same hardware and software platform.

C. Model Analysis

1) Ablation Study: First, ablation experiments are conducted on the image encoder. As shown in Table V, the contribution of ViTs in the image encoder is compared under the scenarios of using text information or not. The model variants include: 1) "Visual(LDGnet)" refers to the image encoder without ViTs, which degrades to the image encoder used in LDGnet [10]; 2) "Visual(Ours)" refers to the image encoder of EHSnet with a serial structure of residual-style CNNs and ViTs with cross-layer connections; and 3) "Text(Ours)" refers to using our proposed multilayered text modal information. Based on the results, it is evident that the introduction of ViTs is beneficial for the model to achieve superior performance, irrespective of the utilization of text information. This validates the effectiveness of the serial structure of the hybrid image encoder employed by EHSnet in extracting the intrinsic spatial–spectral features of HSIs. In addition, the results presented in Table V also demonstrate the effectiveness of our proposed multilayered text modal information.

2) Time Complexity Analysis: To compare the time complexities of different methods, the time consumed to run an epoch on different datasets is presented in Table VI. All experiments are conducted in a consistent environment to ensure fairness. As can be seen, although EHSnet achieves the highest classification accuracy, it has a longer training time compared to other DG methods. This is because EHSnet uses the CLIP-ViT-B-32 pretrained text encoder, and the encoding of two types of texts becomes the main burden of computational cost. Compared with LDGnet, which also uses text information, EHSnet requires less time, mainly because EHSnet fuses two types of text features and only needs to align with the image features once after entering the semantic space.

3) Parameter Sensitivity Analysis: Two parameter sensitivity experiments are conducted for EHSnet. One experiment focuses on the text fusion parameter α, which determines the contributions of global and local semantic information to the model, with a value range of $[0.1, 0.3, 0.5, 0.7, 0.9]$. The other experiment investigates the loss function balancing parameter β, a crucial hyperparameter determining the impact of the supervised contrast loss, with a value range of $[1e-2, 1e-1, 1e+0]$. Tables VII and VIII present the parameter sensitivity experiment results on three datasets. It can be seen that the optimal α values for the Houston, Pavia, and XS datasets are 0.1, 0.3, and 0.7, respectively. The optimal β value for both the Houston and Pavia datasets is $1e+0$, and for the XS dataset is $1e-1$.

In EHSnet, α is a parameter that balances local text with global text. For conventional datasets that lean toward the first-level category, such as Houston and XS, local text plays a more important role. Thus, the ideal value of α for these datasets is between 0.1 and 0.3. In contrast, the Pavia dataset has a more detailed classification, focusing on the classification of roofing and building materials, in which case the performance is more ideal with an α value between 0.5 and 0.7. For datasets with more detailed categories, the proportion of global textual information should be increased. As for β, the balance between the text encoder and the image encoder should be emphasized, as they are paired information. Therefore, the ideal value for β should be 1.0. Compared to other datasets, the XS dataset has a coarser label division and a more fragmented category distribution, which has a stronger visual impact. Hence, the performance is slightly better when β is set to 0.1 than when it is set to 1.0.

D. Quantitative Results and Analysis

The experimental results on the Houston, Pavia, and XS datasets are presented in Tables IX–XI, including the $F1$-score for each class in each dataset, as well as the OA score, AA score, and Kappa coefficient.

From the experimental results, EHSnet outperforms the other eight comparison methods in OA, AA, and Kappa of the three datasets, fully demonstrating the significant advantages of EHSnet. Compared to Meta-causal, which outperforms the five DG methods for natural images, EHSnet has improved OA, AA, and Kappa by about 3%–4%, 1%–8%, and 3%–6% on each dataset, respectively. In comparison with LDGnet, which outperforms the three DG methods for cross-scene HSI classification, EHSnet has improved OA, AA, and Kappa by about 3%–4%, 2%–11%, and 3%–8% on each dataset, respectively. The experimental results prove that EHSnet effectively extracts the intrinsic domain-invariant features of HSIs. The powerful representation capability provided by the proposed multilayer EHS paradigm and the hybrid encoder outperforms methods such as domain expansion and data augmentation. In addition, comparison experiments with LDGnet have proved that the multilayered EHS paradigm proposed by EHSnet is more effective than providing abundant prior knowledge for each HSI category. The complementarity of global and local semantic information significantly enhances EHSnet's learning capability for visual representation. Moreover, the use of prompt templates in EHSnet greatly reduces the generation cost of language modal information.

It can also be observed that EHSnet demonstrates a more significant increase in Kappa across the three datasets than OA. Using LDGnet's experimental results as an example, EHSnet outperforms OA by 4.01%, 2.74%, and 2.88% on Houston, Pavia, and XS datasets, respectively. Correspondingly, Kappa shows improvements of 8.17%, 3.22%, and 3.62%. This superior performance can be attributed to the multilayered EHS information acting as cross-domain shared prior knowledge. This prior knowledge provides consistent semantic information for each category regardless of the sample number. The addition of deep spectral–spatial features extracted by the hybrid encoder enables EHSnet to achieve more balanced classification results when facing datasets with highly imbalanced sample number distributions like the Houston dataset. The results presented in Table IX also corroborate this point. Specifically, it can be observed that the AA score of EHSnet on the Houston dataset exhibits an improvement of 11.02% compared to LDGnet.

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TABLE V
ABLATION RESULTS OF THE IMAGE ENCODER ON THE THREE DATASETS (%)

Visual (LDGnet)	Visual (Ours)	Text (Ours)	Houston		Pavia		XS	
			OA	Kappa	OA	Kappa	OA	Kappa
✓			77.15	60.54	82.39	78.78	63.78	51.61
	✓		78.11	61.88	84.10	80.85	64.81	53.11
✓		✓	79.40	65.55	83.54	80.07	64.61	52.98
	✓	✓	**81.60**	**68.50**	**86.13**	**83.30**	**67.04**	**55.89**

TABLE VI
DURATION OF TRAINING AN EPOCH ON THREE DATASETS USING VARIOUS METHODOLOGIES (SECONDS)

dataset	PDEN [27]	SagNet [31]	Meta-causal [32]	FedDG-DA [33]	ABA [28]	SDEnet [30]	LLURnet [29]	LDGnet [10]	Ours
Houston	3.17	1.83	9.03	16.80	121.66	6.28	10.59	16.21	13.44
Pavia	4.25	5.36	34.73	34.00	122.20	20.24	18.10	61.65	52.37
XS	5.98	14.84	49.08	80.45	130.08	57.76	48.50	153.44	147.96

TABLE VII
OVERALL ACCURACY (%) AND KAPPA (%) WITH DIFFERENT BALANCING PARAMETER α FOR THE PROPOSED EHSnet ON THE THREE DATASETS

α	Houston		Pavia		XS	
	OA	Kappa	OA	Kappa	OA	Kappa
0.1	**81.60**	**68.50**	81.75	78.03	65.93	54.61
0.3	79.26	64.38	82.31	78.71	**67.04**	**55.89**
0.5	79.28	64.31	83.16	79.59	66.07	54.89
0.7	78.52	64.23	**86.13**	**83.30**	65.80	54.59
0.9	77.23	61.41	83.07	79.58	66.02	54.78

TABLE VIII
OVERALL ACCURACY (%) AND KAPPA (%) WITH DIFFERENT BALANCING PARAMETER β FOR THE PROPOSED EHSnet ON THE THREE DATASETS

β	Houston		Pavia		XS	
	OA	Kappa	OA	Kappa	OA	Kappa
1e-2	76.42	58.61	79.19	74.98	66.10	54.83
1e-1	79.80	64.56	83.74	80.53	**67.04**	**55.89**
1e+0	**81.60**	**68.50**	**86.13**	**83.30**	66.81	55.58

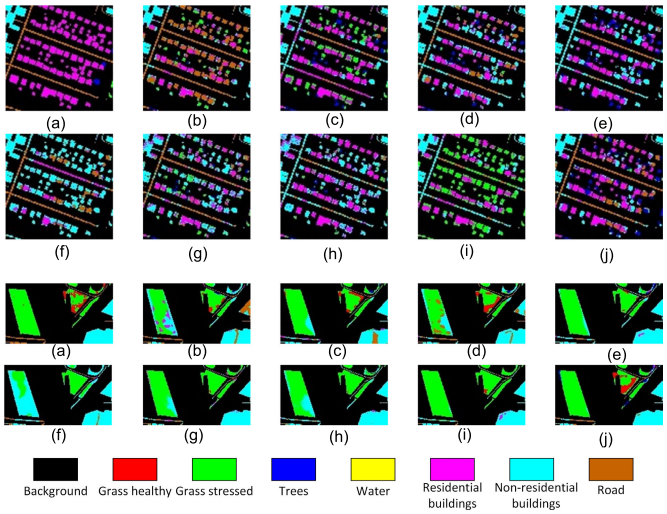


Fig. 6. Visualization of classification results from local regions in the Houston dataset predicted by different methods. (a) Truth map. (b) PDEN. (c) SagNet. (d) Meta-causal. (e) FedDG-DA. (f) ABA. (g) SDEnet. (h) LLURnet. (i) LDGnet. (j) Ours.

E. Visual Evaluation

Figs. 6–8 provide the visualization of the classification results in the local regions of the three datasets using different methods. It can be seen that EHSnet has achieved better classification results and less noise in these areas. For example, EHSnet has better distinguished between residential buildings and nonresidential buildings in the Houston dataset, asphalt and bitumen in the Pavia dataset, and orchard and tree in the XS dataset.

Fig. 9 provides the visualization of class separability before and after the inference of the EHSnet following t-SNE with the Pavia datasets. It can be seen that EHSnet has significantly improved class separability and the aggregate of the same class is higher. For instance, the aggregate of category-4 (Shadow) is significantly enhanced after the inference of the trained EHSnet, reflected in an impressive $F1$-score of 97.5% for category-4 (Shadow) as shown in Table X.

F. Effects of the Multilayer EHS Structure

Ablation experiments are conducted on the image encoder of EHSnet and the multilayered EHS paradigm to investigate their contributions to the model.

1) Advantages of the Multilayer EHS: To investigate the performance gain of the proposed multilayer EHS module in terms of classification accuracies, ablation studies are conducted involving the removal or combination of key components: 1) removing patch text in EHSnet; 2) removing global text in EHSnet; and 3) combining patch text and global text to be sent directly to the text encoder. For that, we conduct extensive ablation experiments on the three datasets to verify the effectiveness and advantages of multilayer EHS, as listed in Table XII.

In detail, the combination of patch text and global text sent directly to the text encoder achieves the lowest accuracy, even slightly lower than the performance of the proposed image encoder alone (see Table V). The accuracies obtained from using single-layer text, such as removing patch text or global text, show similar performance. However, when the global text and patch text are structured together to form the multilayer EHS, the accuracy reaches the highest level. For example, on the Houston dataset, multilayer EHS surpasses the accuracy of the single-layer global text prompt by 3.06% in OA and

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TABLE IX
PERFORMANCE ON THE HOUSTON DATASET USING HOUSTON 2013 AS THE SOURCE DOMAIN AND HOUSTON 2018 AS THE TARGET DOMAIN (%)

NO	Class	PDEN [27]	SagNet [31]	Meta-causal [32]	FedDG-DA [33]	ABA [28]	SDEnet [30]	LLURNet [29]	LDGnet [10]	Ours
1	Grass healthy	44.08	58.11	65.97	34.16	07.23	29.92	38.87	44.62	56.72
2	Grass stressed	57.42	66.11	64.40	75.86	42.26	66.15	69.58	68.34	78.76
3	Trees	65.68	63.48	63.80	62.74	22.93	66.97	70.11	60.67	57.70
4	Water	17.19	50.00	74.58	80.00	74.29	20.75	48.89	58.06	32.06
5	Residential buildings	51.71	53.54	65.06	54.47	42.02	46.90	56.73	62.42	66.71
6	Non-residential buildings	79.41	82.74	86.64	85.84	80.89	86.00	83.49	88.96	90.06
7	Road	53.88	36.20	62.11	41.63	46.26	63.47	55.31	34.43	68.56
	OA	68.78	70.70	78.04	75.99	68.71	74.58	74.63	77.59	**81.60**
	AA	64.24	64.65	71.41	62.72	39.28	65.75	66.01	61.56	**72.58**
	Kappa	50.20	52.95	62.05	55.97	36.60	58.31	56.23	60.33	**68.50**

TABLE X
PERFORMANCE ON THE PAVIA DATASET USING PAVIA UNIVERSITY AS THE SOURCE DOMAIN AND PAVIA CENTER AS THE TARGET DOMAIN (%)

NO	Class	PDEN [27]	SagNet [31]	Meta-causal [32]	FedDG-DA [33]	ABA [28]	SDEnet [30]	LLURNet [29]	LDGnet [10]	Ours
1	Tree	86.29	82.09	84.57	92.20	86.14	86.60	87.26	88.25	90.55
2	Asphalt	79.52	83.08	82.12	79.32	73.70	84.84	84.85	87.38	87.71
3	Brick	64.23	08.73	38.55	24.50	10.86	60.63	56.31	67.31	75.51
4	Bitumen	77.64	74.69	85.52	71.28	74.05	82.28	79.00	85.72	85.86
5	Shadow	86.30	87.97	91.94	94.05	76.03	90.68	92.10	86.83	97.57
6	Meadow	65.05	65.29	67.86	51.69	68.13	66.72	66.39	71.54	68.84
7	Bare soil	79.38	83.59	89.94	65.21	84.67	82.54	79.98	82.19	86.74
	OA	78.14	77.02	81.72	73.85	75.33	81.03	80.49	83.46	**86.13**
	AA	78.95	70.72	77.01	69.43	67.83	80.73	78.05	81.68	**85.14**
	Kappa	73.86	72.11	77.85	68.42	69.90	77.31	76.49	80.08	**83.30**

TABLE XI
PERFORMANCE ON THE XS DATASET USING XS_0 AS THE SOURCE DOMAIN AND XS_1 AS THE TARGET DOMAIN (%)

NO	Class	PDEN [27]	SagNet [31]	Meta-causal [32]	FedDG-DA [33]	ABA [28]	SDEnet [30]	LLURNet [29]	LDGnet [10]	Ours
1	Road	0	0	27.61	34.39	01.70	21.06	24.43	21.74	20.08
2	Building	81.83	80.86	83.34	82.73	83.41	82.55	82.51	82.45	83.74
3	Tree	17.56	37.47	34.96	35.26	25.42	28.63	31.61	42.34	52.40
4	Farmland	37.73	0	29.38	31.86	30.99	31.06	30.52	26.50	22.32
5	Bare Land	55.88	65.22	62.63	59.62	64.22	59.32	57.76	63.02	65.46
6	Orchard	64.89	61.67	65.58	66.56	65.38	64.28	64.63	66.12	71.84
7	Water	30.15	0	35.15	30.36	32.32	34.13	29.81	36.59	27.82
	OA	59.95	62.39	63.60	62.14	63.84	61.40	61.04	64.16	**67.04**
	AA	43.75	36.62	47.81	47.48	43.70	45.62	45.38	46.67	**48.62**
	Kappa	47.62	48.42	52.07	50.59	51.63	49.30	48.91	52.27	**55.89**

TABLE XII
ABLATION RESULTS OF THE **TEXT STRUCTURE** ON THE THREE DATASETS (%)

Global Text	Patch Text	Combination Text	Houston		Pavia		XS	
			OA	Kappa	OA	Kappa	OA	Kappa
✓			78.54	64.18	84.64	81.48	65.21	53.41
	✓		78.93	62.60	85.56	82.57	65.34	53.68
		✓	77.63	62.06	84.02	80.80	64.18	52.24
✓	✓		**81.60**	**68.50**	**86.13**	**83.30**	**67.04**	**55.89**

4.32% in Kappa. In addition, it exceeds the performance of the combination text by 3.97% in OA and 6.44% in Kappa.

Interestingly, the results suggest that the multilayered EHS, with its finer-grained representation, is more advantageous for the visual–language model. The combining of two strong representations sent directly to the text encoder performs slightly lower than when solely using the image encoder. This may be attributed to the concatenation of the two types of text, which can introduce ambiguities or contradictions in contextual semantics, leading to confusion among different classes for the model. In contrast, multilayer EHS allows for better integration of the various high-level semantic information provided by each text type, thereby enhancing the text encoder's understanding capabilities in a more streamlined and efficient manner. Our findings are consistent with the conclusions of some recent research on EHS.

Notably, in Table V, when classification is performed using only the image encoder, that is, when the class labels are converted to a "one-hot" format for classification, the accuracy on the Pavia dataset is 84.10%. In comparison, when

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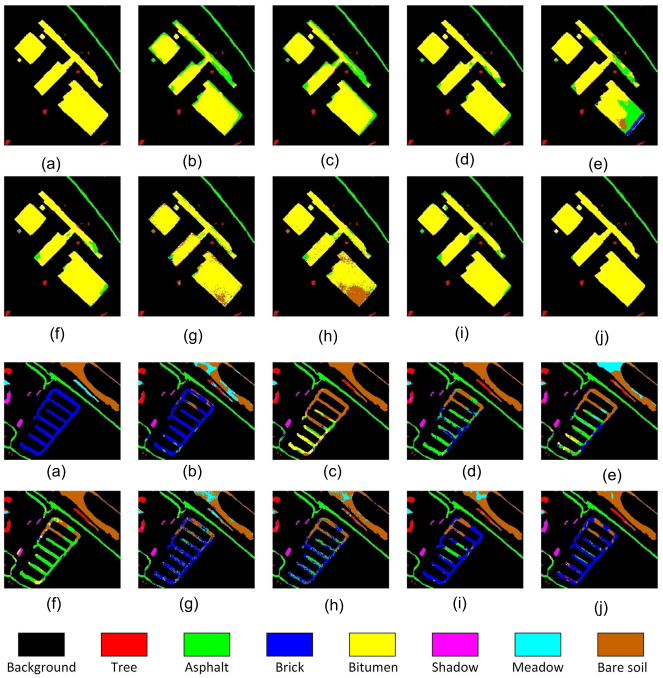


Fig. 7. Visualization of classification results from local regions in the Pavia dataset predicted by different methods. (a) Truth map. (b) PDEN. (c) SagNet. (d) Meta-causal. (e) FedDG-DA. (f) ABA. (g) SDEnet. (h) LLUR-net. (i) LDGnet. (j) Ours.

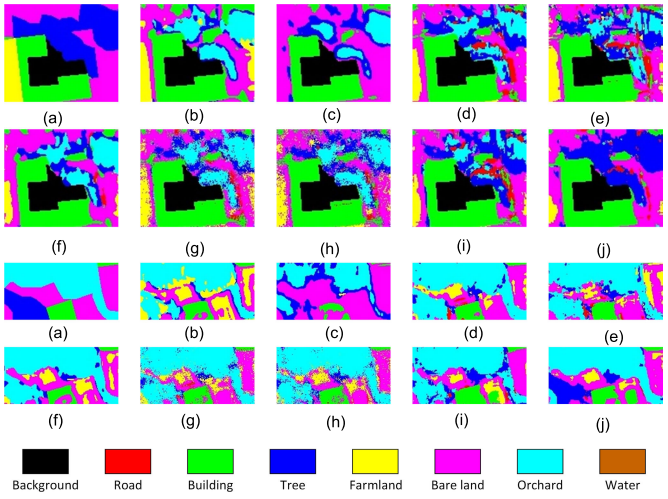


Fig. 8. Visualization of classification results from local regions in the XS dataset predicted by different methods. (a) Truth map. (b) PDEN. (c) SagNet. (d) Meta-causal. (e) FedDG-DA. (f) ABA. (g) SDEnet. (h) LLUR-net. (i) LDGnet. (j) Ours.

a single layer of local language description is employed (only using class labels), it achieves an overall accuracy (OA) of 85.56%, surpassing the "one-hot" method by 1.46% (Table XII). Similar results were found in the other two datasets. The results indicate that the textual encoder with language description can provide richer and more detailed semantic information, whereas one-hot labels can only indicate the presence or absence of categories. Furthermore, language descriptions can incorporate semantic information in more complex forms (such as multilayer EHS), which also have better extensibility. In addition, the textual encoder of language

TABLE XIII
MISINFORMATION GLOBAL TEXT OF TABLE V ON THE THREE DATASETS

dataset	Misinformation global text
Houston	Love is the one thing that transcends time and space.
Pavia	There are only the pursued, the pursuing, the busy and the tired.
XS	Life was a box of chocolates, you never know what you are gonna get.

descriptions can integrate image information from HSIs and semantic information from large language models, thereby obtaining rich information from multiple channels.

2) Comparison With Other Text Information Structures: Next, to illustrate the impact of the proposed multilayered EHS paradigm, ablation experiments are also conducted on different text information structures. Under the same image encoder, the text information variants include: 1) "Global Text" refers to using only global semantic information; 2) "Patch Text" indicates training solely with patch text generated from the prompt template; 3) "Fine-grained Text" refers to the fine-grained text designed in LDGnet [10], which contains category-related prior knowledge; and 4) "Misinformation Global Text" involves training with global information that is completely irrelevant to the datasets (Table XIII). This aims to examine the contribution of the global semantic information provided by the paired global text.

As shown in Table XIV, the optimal performance is attained through our proposed multilayered EHS paradigm. In addition, it is worth to note the following.

1) Using only the patch text generated from the prompt template is insufficient to improve model performance significantly. Using only the semantic information provided by the global text, which is adapted to the entire HSI, cannot help the model distinguish between different categories but instead leads to a decrease in model performance.
2) Our proposed multilayered EHS paradigm has achieved the best performance. This proves that compared to providing additional semantic information for a specific class by combining prior knowledge, the hierarchical semantic structure proposed by EHSnet is more applicable to HSIs. It not only helps the model to achieve better generalization but also reduces the training costs.
3) The experimental results prove that global text mismatching with the dataset negatively affects model performance, indicating the importance of the semantic information the well-designed global text provides. Although EHSnet generates full-image description text through a simple prompt template, it supplements the model's required global semantic information.

3) Design of the Multilevel Prompt Template: To further explain the design of the multilayer prompt templates in this study, comparison experiments were conducted to evaluate the impact of different multilayer prompt templates. For the patch prompt template, we designed two prompt templates: 1) "an image of [class]" and 2) "an HSI of [class]". For the

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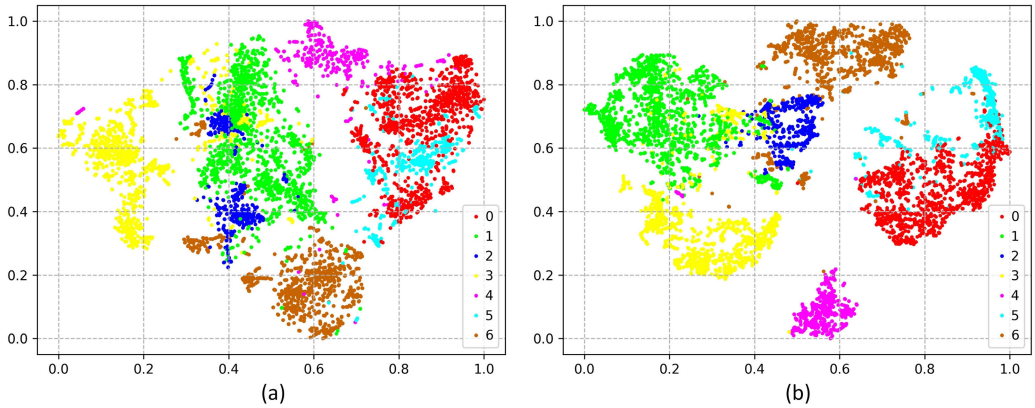


Fig. 9. Class separability of the EHSnet using the Pavia dataset. The source domain refers to Pavia University, the target domain refers to Pavia Center, and the number indicates the class index. (a) Original target-domain sample distribution. (b) Feature distribution after the inference of the trained EHSnet.

TABLE XIV
ABLATION RESULTS OF THE **TEXT INFORMATION** ON THE THREE DATASETS (%)

Global Text	Patch Text	Fine-grained Text	Misinformation Global Text	Houston		Pavia		XS	
				OA	Kappa	OA	Kappa	OA	Kappa
✓				78.54	64.18	84.64	81.48	65.21	53.41
	✓			78.93	62.60	85.56	82.57	65.34	53.68
	✓	✓		77.64	61.58	84.84	81.72	64.75	53.49
	✓		✓	75.73	60.13	81.50	77.74	63.01	50.99
✓	✓			**81.60**	**68.50**	**86.13**	**83.30**	**67.04**	**55.89**

TABLE XV
ABLATION RESULTS OF THE **TEXT PROMPT TEMPLATE** ON THE THREE DATASETS (%)

Patch Text	Global Text	Houston		Pavia		XS	
		OA	Kappa	OA	Kappa	OA	Kappa
p1	g3	80.58	65.44	85.71	81.45	66.27	54.52
p2	g3	**81.60**	**68.50**	**86.13**	**83.30**	**67.04**	**55.89**
p2	g1	80.04	63.21	85.54	82.02	66.44	54.61
p2	g2	80.86	65.05	85.64	81.97	66.14	54.64
p2	g3	**81.60**	**68.50**	**86.13**	**83.30**	**67.04**	**55.89**

patch text prompt template
p1: 'An image of [$class$]'
p2: 'A hyperspectral image of [$class$]'
global text prompt template
g1: 'A hyperspectral image of [$class_1$]...[$class_n$]'
g2: 'A hyperspectral image with surface covered with [$class_1$]...[$class_n$]'
g3: 'A hyperspectral image of [$simple\ description$] with surface covered with [$class_1$]...[$class_n$]'

global text, we designed three templates: 1) "an HSI of [class$_1$]...[class$_n$];" 2) "an HSI with a surface covered with [class$_1$]...[class$_n$];" and 3) "an HSI of [simple description] with a surface covered with [class$_1$]...[class$_n$]." For that, we conduct extensive comparison experiments on the three datasets to verify the effectiveness and advantages of the multilayer EHS design in this study, as listed in Table XV.

In detail, the local text prompt template, by incorporating the simple prompt "hyperspectral" and modifying it from "an image of [class]" to "an HSI of [class]," results in improved performance across all three datasets (Table XV). Similarly, for the global prompt template, the addition of variable descriptions such as [simple description] alongside fixed descriptions like "with surface covered" also contributes to enhanced model performance on the three datasets (Table XV).

Previous ablation studies have demonstrated that imposing the integration of information from different layers does not promote an effective understanding of the model. In contrast, utilizing simple prompts across various layers has been shown to better guide the EHS performance. Consequently, we designed the patch text prompt template as "an HSI of [label]." To enrich the semantic content, we developed a global text template that incorporates straightforward details about the image, including location information. This approach allows for a simpler integration of foundational information into the global text compared to the addition of class-related information in the patch text. Thus, the global text template is articulated as "an HSI of [simple description] with a surface covered with [label$_1$]...[label$_n$]." The comparative results demonstrate the superior performance of this multilayer prompt template design.

V. CONCLUSION

This article proposes a novel cross-scene classification method for HSIs, EHSnet. Specifically, a hybrid image encoder

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with a serial structure of residual-style CNNs and ViTs with cross-layer connections is built to extract the intrinsic spatial–spectral features of HSIs. Moreover, EHSnet proposes a multilayered EHS paradigm that integrates image-level and patch-level text information. Through the fusion of the two types of semantic information and the application of supervised contrastive learning, the image encoder demonstrates strong representation capability in learning class-specific domain-invariant features. This provides EHSnet with great zero-shot generalization ability for cross-scene tasks. Extensive experiments conducted on three different HSI datasets prove that the performance of EHSnet in the cross-scene classification of HSIs is superior to the current SOTA methods. Ablation experiments show that the hybrid image encoder and multilayered EHS paradigm we proposed significantly enhance the representation capability of EHSnet, achieving better generalization.

In this study, the datasets used by EHSnet and other cross-scene HSI classification methods need to be preprocessed to make the source and target domains share the same categories and maintain a consistent label–index correspondence relationship, limiting the model's further application. In future work, we will concentrate on helping the model overcome this limitation, thus enhancing its generalization ability.

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